



UNMASKING FRAUD: WHAT TRULY MATTERS?

UNIVERSITY OF MICHIGAN

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Authors:

Jake Gwinn

Jiacheng You

Huan Wang

Li Wang

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1 Summary

Our project explores the underlying determinants of financial fraud by analyzing transaction records collected between 2017 and 2019. To address the complexities in fraud detection, we compare the performance of a baseline Generalized Linear Model against more advanced Generalized Additive Models and Generalized Additive Mixed Models. Our methodology incorporates a rigorous numerical simulation study to evaluate these models across different scenarios, utilizing AUC and Brier scores to assess their explanatory power and predictive accuracy. The results demonstrate that the GAMM offers best performance by capturing both non-linear trends and unobserved heterogeneity across industries and geographic regions. Additionally, we identify specific risk factors—including seasonal spikes, high credit card ownership, and specific card brands—confirming that accurate fraud unmasking requires a modeling framework capable of accommodating multi-level data structures.

2 Introduction

Financial fraud remains one of the most persistent risks faced by modern banking institutions. As electronic payment systems expand and transaction volumes grow, identifying fraudulent behavior quickly and accurately has become essential to minimize financial loss and protect customers. The data set used in this study integrates transaction records, demographic information from customers, and card-level attributes collected by a financial institution between 2017 and 2019. This data provides a comprehensive view of user behavior, spending patterns, and contextual information surrounding each transaction.

Fraudulent activities are rare events in real-world financial systems, yet they carry a disproportionate impact. Understanding the conditions under which fraud is more likely to occur can help financial institutions develop more robust fraud detection systems, strengthen risk management protocols, and design monitoring strategies that adapt to evolving patterns of misuse. This project builds on these motivations by using statistical modeling and exploratory data analysis to uncover the features most associated with fraudulent transactions.

We will use a GLM model as the baseline and investigate potential non-linear relationships and random effects through more advanced GAM and GAMM frameworks. We will compare the performance of these models and select the one that offers the strongest explanatory power.

3 Data

3.1 Data Transformations

Our dataset contains transaction-level observations with a binary response variable indicating whether a transaction is fraudulent. The raw data includes customer demographics, credit attributes, merchant identifiers, temporal information, and categorical indicators such as card type, card brand, and transaction location. Fraud transactions are rare in this dataset. The overall fraud rate (`is_fraud = 1`) is approximately 1.4%, indicating a strong class imbalance. This characteristic affects both model training and evaluation metrics, and motivates several additional preprocessing steps.

Before model building, we applied several preprocessing steps to ensure data quality and statistical validity.

Conducting VIF and Prior Power Analysis We first examined multicollinearity among continuous predictors using Variance Inflation Factors (VIF). Most variables showed VIF values below commonly accepted thresholds, indicating no severe multicollinearity. And we keep one of variables who show multicollinearity. In addition, we performed a prior power analysis to assess whether the sample size was adequate for detecting fraud-related effects. Given the rarity of fraud, evaluating statistical power is essential for ensuring that model estimates are stable and interpretable.

Filtering In-State vs. Out-of-State Transactions The dataset reveals a striking pattern: all fraud observations occur in interstate transactions. Because in-state transactions contain no fraud cases, they provide no discriminatory value for learning fraud patterns. Therefore, we drop in-state transactions to avoid inflating the majority class and to focus model training on the portion of the dataset relevant to fraud detection.

Feature Engineering We generated several new variables to enhance predictive performance. For example, `account_age`: computed as the number of years since the user's account was opened. Additional engineered features consider temporal spacing, usage behavior, and transformations of existing financial attributes. These engineered features help uncover nonlinear relationships and improve model expressiveness.

Log Transformations of Skewed Predictors Many continuous variables, especially amount-related and time-related measures, exhibit substantial variation and right-skewed distributions. To stabilize variance, reduce skewness, and help linear models fit more effectively, we applied log transformations. Log-transformed predictors show improved distributional properties and reduce the influence of extreme values.

Table 1 summarizes key categorical variables in the dataset and describes how they relate to fraud occurrence. For each variable, we report the number of unique levels, along with the mean, standard deviation, and maximum

observed fraud rate across those levels. These statistics help identify which categorical features exhibit substantial variation in fraud risk. Variables such as *merchant_id* and *mcc* have a large number of levels and show high maximum fraud rates, suggesting strong heterogeneity across merchants and merchant categories. In contrast, variables like *time_of_day* and *season* exhibit much lower variability, indicating more consistent fraud patterns across their categories.

Table 1: Categorical Variables in Relation to Fraud Rate

Variable	Num_Levels	Mean_Fraud_Rate	SD_Fraud_Rate	Max_Fraud_Rate
merchant_id	165	0.0355	0.1446	0.9780
mcc	53	0.0331	0.1189	0.8348
time_of_day	4	0.0118	0.0114	0.0261
state_abbr	49	0.0155	0.0077	0.0455
month	12	0.0149	0.0040	0.0238
card_brand	4	0.0163	0.0040	0.0222
season	4	0.0148	0.0030	0.0170
card_type	2	0.0150	0.0027	0.0170

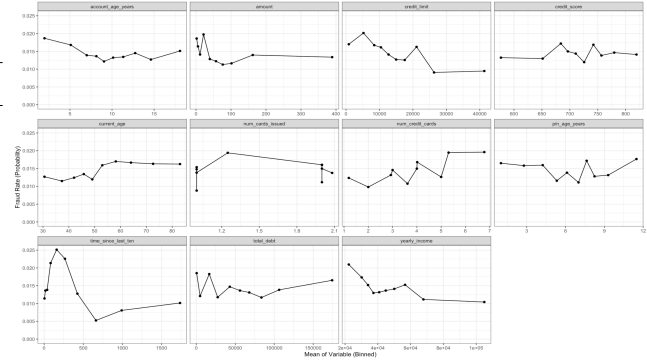


Figure 1: Relationship Between Continuous Predictors and Fraud Rate

Figure 1 illustrates how fraud rate varies across binned values of several continuous predictors. Overall, the patterns indicate that fraud risk does not increase monotonically with most financial or demographic attributes. Instead, several variables show nonlinear relationships. For example, the fraud rate tends to decrease for higher account age and larger credit limits, while variables such as time since last transaction display a sharp peak in fraud risk at shorter intervals. With this information, we should consider non-linear trends.

4 Method

4.1 Generalized Linear Model

To fully understand our data, we first wanted to fit a baseline model. We chose to fit a Generalized Linear Model (GLM) using a logit link function. We assume that our response variable y_i for $i = 1, \dots, n$ which takes the values zero or one with $P(y_i = 1) = p_i$. We then construct a linear predictor with our k predictors^[1]. Formally it is as follows:

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik}$$

Since we require p_i to be between zero and one, we need to convert η_i to that same scale. We use the logit function defined as $\log(p/(1-p))$. Combining this logit link with our linear predictors allows us to conduct logistic regression.

Therefore for our GLM fraud-risk model, it is fully defined as:

$$\begin{aligned}
\text{logit}(P(Y_i = 1)) = & \beta_0 + \beta_1 \log_amount_i + \beta_2 \log_time_since_i + \beta_3 \log_income_i + \beta_4 \log_total_debt_i \\
& + \beta_5 \text{credit_score}_i + \beta_6 \text{num_credit_cards}_i + \beta_7 \text{num_cards_issued}_i + \beta_8 \log_credit_limit_i \\
& + \sum_{j=1}^{K_1-1} \beta_j^{(\text{season})} \text{season}_{ij} + \sum_{j=1}^{K_2-1} \beta_j^{(\text{gender})} \text{gender}_{ij} + \sum_{j=1}^{K_3-1} \beta_j^{(\text{brand})} \text{card_brand}_{ij} + \sum_{j=1}^{K_4-1} \beta_j^{(\text{type})} \text{card_type}_{ij} \\
& + \sum_{j=1}^{K_5-1} \beta_j^{(\text{time})} \text{time_of_day}_{ij}.
\end{aligned}$$

This will allow us to get an effective baseline result to compare are other models to as this is a fairly simplistic model.

4.2 Generalized Additive Model

To capture nonlinear relationships between the covariates and the probability of fraud risk, we employed a logistic Generalized Additive Model (GAM) [4, 6]. A GAM extends the classical generalized linear model by replacing certain linear terms with smooth functions estimated via penalized splines. For the fraud-risk application, the model is specified as:

$$\text{logit}\{P(\text{fraud})\} = \beta_0 + \sum_{k=1}^K \beta_k Z_k + f_1(\log_amount) + f_2(\log_income) + f_3(\log_total_debt) + f_4(\log_credit_limit) + f_5(\log_time_since)$$

GAM is suitable here because of the following: It flexibly models nonlinearities without specifying a parametric form; It provides interpretable smooth terms and uncertainty intervals; Estimation via penalized likelihood with REML is computationally stable; The method is straightforward to implement using the `mgcv` package. Estimation follows the penalized-likelihood framework of wood[6], where smoothing parameters are selected via REML. This approach balances bias and variance while preventing oversmoothing or overfitting.

4.3 Generalized Additive Mixed Model

A Generalized Additive Mixed Model (GAMM) augments the additive predictor with group-specific random terms:

$$\text{logit}\{P(\text{fraud})\} = \beta_0 + \sum_{k=1}^K \beta_k Z_k + \sum_{j=1}^5 f_j(X_j) + b_{\text{merchant}}^{(\text{merch})} + b_{\text{mcc}}^{(\text{mcc})} + b_{\text{state}}^{(\text{state})},$$

where Z_k are our linear predictors (credit_score, num_credit_cards, num_cards_issued, current_age, account_age_years, pin_age_years, gender, card_brand, card_type, season, time_of_day), X_j are the continuous covariates with smooth terms (log_amount, log_income, log_total_debt, log_time_since, log_credit_limit), and

$$b_{\text{merchant}}^{(\text{merch})} \sim N(0, \sigma_{\text{merch}}^2), \quad b_{\text{mcc}}^{(\text{mcc})} \sim N(0, \sigma_{\text{mcc}}^2), \quad b_{\text{state}}^{(\text{state})} \sim N(0, \sigma_{\text{state}}^2)$$

capture unobserved heterogeneity at the merchant, MCC, and state levels.

In this analysis, GAMM is motivated by: Residual patterns from the baseline GAM indicating unresolved group-

level variation; Large within-group variance across merchant ID, MCC, and geographic units; The need to separate nonlinear continuous effects from discrete; unobserved noise sources.

Our GAMM is estimated via REML using `mgcv::gamm`, enabling posterior inference on smooth terms and random effects. GAMM should improve predictive performance, reduce overdispersion, and provide a more coherent decomposition of nonlinear and group-level structure.

5 Analysis

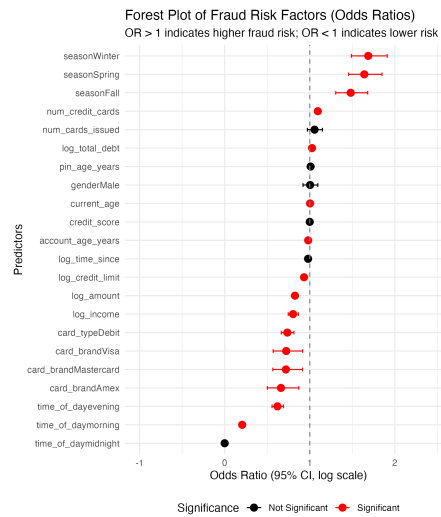
5.1 Generalized Linear Model

To understand our more complex models and how they adapt differently to the data, it is important that we do some initial analysis and interpretation of our results from our generalized linear model. Looking at Figure 2, we find that the only non-significant terms are `num_cards_issued`, `pin_age_years`, `log_time_since`, and `time_of_daymidnight`. Some initial interpretations of significant coefficients are that later seasons like Winter and Fall increase the log odds of fraud as compared to summer. This could be due to more holidays being during these times (Black Friday, Christmas, etc.) and so there are a lot more transactions and therefore, more opportunities for fraudsters to strike. Alternatively we see that the card type being debit, log income, and log amount all decrease the log odds of fraud. For card type, people might be more cautious of their debit card as it is directly linked to their bank account. For amount, we see that larger purchases are less likely to be fraud indicating that fraudsters would rather try and just get a smaller amount than risk trying to buy something very large.

Table 2: Coefficient summary from the GLM Model

Term	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.25	0.51	-0.50	0.62
log_amount	-0.19	0.02	-11.76	0.00
log_time_since	-0.02	0.01	-1.59	0.11
log_income	-0.22	0.04	-5.79	0.00
log_total_debt	0.03	0.01	2.84	0.00
credit_score	-0.00	0.00	-0.93	0.35
num_credit_cards	0.09	0.02	5.90	0.00
num_cards_issued	0.06	0.04	1.29	0.20
log_credit_limit	-0.07	0.01	-5.25	0.00
seasonFall	0.39	0.06	6.09	0.00
seasonSpring	0.50	0.06	8.13	0.00
seasonWinter	0.52	0.06	8.25	0.00
current_age	0.00	0.00	2.17	0.03
account_age_years	-0.02	0.01	-3.35	0.00
pin_age_years	0.01	0.01	1.24	0.22
genderMale	0.00	0.04	0.09	0.93
card_brandAmex	-0.41	0.14	-2.95	0.00
card_brandMastercard	-0.33	0.12	-2.69	0.01
card_brandVisa	-0.32	0.12	-2.68	0.01
card_typeDebit	-0.31	0.05	-6.01	0.00
time_of_dayevening	-0.48	0.06	-8.65	0.00
time_of_daymidnight	-15.93	109.94	-0.14	0.88
time_of_daymorning	-1.58	0.06	-25.48	0.00

Figure 2: Coefficient Significance Plot



Investigating further with Table 2, we find that `time_of_daymidnight` has an extremely large standard error. We found this is due to there being no fraud instances during this time frame and thus should be interpreted with caution. We also found that our model had moderate to large over-dispersion value of 1.92. This indicates that all sources of variability are not being captured. This is most likely due to our model missing the group-like structure of our data (clients and merchants having multiple transactions) that we hope to capture in subsequent models.

5.2 Generalized Additive Model

Overall, clear nonlinear patterns emerge for transaction amount, total debt and recency: fraud risk increases for large transaction amounts and high debt levels, with some flattening or threshold behavior at the extremes. These patterns cannot be captured by the linear GLM and further motivate the use of GAMs.

Table 3: Parametric Coefficients from the GAM Model

Term	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.4340	0.4142	-10.7040	0.00
credit_score	0.0000	0.0004	0.0700	0.94
num_credit_cards	0.0828	0.0160	5.1640	0.00
num_cards_issued	0.0844	0.0432	1.9540	0.05
current_age	0.0035	0.0021	1.6980	0.09
account_age_years	-0.0172	0.0058	-2.9590	0.00
pin_age_years	0.0158	0.0083	1.9110	0.06
genderMale	-0.0210	0.0462	-0.4540	0.65
card_brandAmex	-0.4325	0.1427	-3.0310	0.00
card_brandMastercard	-0.3907	0.1244	-3.1400	0.00
card_brandVisa	-0.4022	0.1228	-3.2750	0.00
card_typeDebit	-0.2757	0.0521	-5.2940	0.00
seasonFall	0.4190	0.0652	6.4260	0.00
seasonSpring	0.5195	0.0616	8.4300	0.00
seasonWinter	0.5106	0.0643	7.9460	0.00
time_of_dayevening	-0.4863	0.0569	-8.5430	0.00
time_of_daymidnight	-13.7600	39.6700	-0.3470	0.73
time_of_daymorning	-1.4830	0.0651	-22.8020	0.00

From Table 3, we find that most effects are highly significant. Holding other variables fixed, having more credit cards and more cards issued is associated with a higher probability of fraud, whereas longer account age reduces risk. *Discover* cards and winter/spring seasons substantially increase fraud odds, while debit cards and evening transactions are associated with lower fraud risk as compared to noon.

Finally, based on the smooth-term diagnostics produced by our model, all spline components show extremely strong significance, with large Chi-square statistics and p-values below $2e-16$. This indicates that nonlinear effects in key financial variables are essential for explaining fraud risk.

5.3 Generalized Additive Mixed Model

Interpretation of Significant Parametric Effects The significant parametric terms of the GAMM highlight several key determinants of fraud risk after accounting for nonlinear smooth components and merchant-id, MCC-id, and

state-level random effects.

Table 4: Parametric Coefficients from the GAMM Model

Term	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-7.7340	0.7132	-10.8430	0.00
credit_score	0.0004	0.0005	0.8040	0.42
num_credit_cards	0.0940	0.0228	4.1320	0.00
num_cards_issued	0.0802	0.0579	1.3840	0.17
current_age	0.0010	0.0028	0.3460	0.73
account_age_years	-0.0056	0.0075	-0.7390	0.46
pin_age_years	0.0021	0.0112	0.1830	0.85
genderMale	0.0849	0.0627	1.3540	0.18
card_brandAmex	-0.5848	0.1889	-3.0960	0.00
card_brandMastercard	-0.4714	0.1646	-2.8650	0.00
card_brandVisa	-0.4254	0.1623	-2.6210	0.01
card_typeDebit	-0.2734	0.0705	-3.8800	0.00
seasonFall	0.4919	0.0849	5.7930	0.00
seasonSpring	0.4765	0.0809	5.8930	0.00
seasonWinter	0.4682	0.0848	5.5230	0.00
time_of_dayevening	-0.3419	0.0757	-4.5150	0.00
time_of_daymidnight	-22.6200	2997.0000	-0.0080	0.99
time_of_daymorning	-1.3030	0.0813	-16.0310	0.00

Card-related attributes are influential: the number of credit cards (`num_credit_cards`) shows a strong positive association with fraud likelihood, suggesting that customers managing multiple credit lines face greater exposure. In contrast, debit card usage (`card_typeDebit`) is linked to substantially lower fraud odds, consistent with more routine and lower-value spending patterns. Brand differences also emerge. *Discover* transactions exhibit significantly higher fraud risk relative to the baseline *Amex*, reflecting heterogeneous fraud environments shaped by user demographics, merchant acceptance, or historical targeting patterns.

Temporal factors further differentiate risk. Fraud risk increases in Fall, Spring, and Winter (compared to the baseline of summer) indicating seasonality aligned with higher shopping volumes and promotional activity. Time-of-day effects show that morning transactions carry significantly elevated risk, whereas evening activity is associated with reduced odds.

Overall, these effects demonstrate that fraud risk arises from a combination of card characteristics, seasonal dynamics, and behavioral timing patterns, with random intercepts ensuring that these fixed-effect interpretations reflect genuine individual-level behavior rather than group-level confounding.

Interpretation of Smooth Terms **Log Amount (Figure 3)** The smooth term for `log_amount` displays a clear U-shaped pattern. Fraud risk is elevated for very small transactions, consistent with a “probe”-like behavior in which stolen cards are tested using low-value charges. The effect declines and stabilizes across moderate transaction sizes, suggesting typical spending amounts are not strongly linked to fraud. At higher values ($\log > 6$), the curve rises again, indicating that unusually large transactions are also associated with increased fraud likelihood. Thus, both extremely small and extremely large amounts show disproportionate risk.

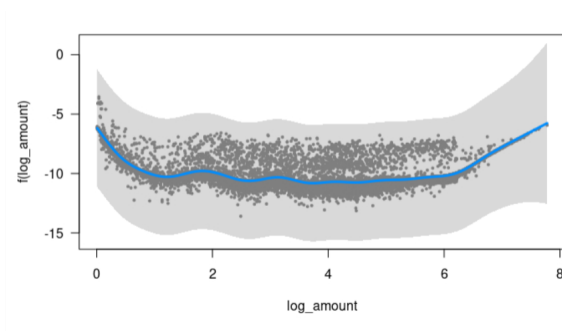


Figure 3: Log Amount Distribution

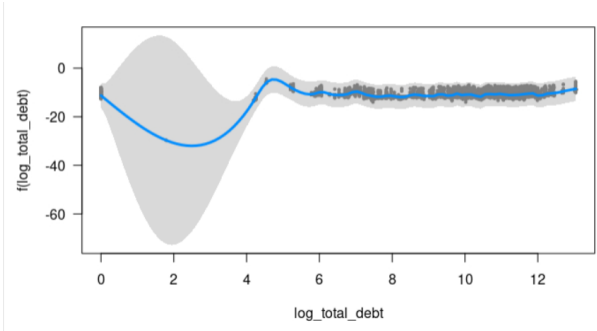


Figure 4: Log Total Debt Distribution

Log Total Debt (Figure 4) The smooth for `log_total_debt` shows mild non-linearity, but much of the high-debt variability is absorbed by the GAMM's random effects, producing an almost flat trend in the upper range. The strong curvature at very low debt levels should be interpreted cautiously due to data sparsity and wide uncertainty bands. The mid-range ($\log(2)$ – $\log(6)$) provides the most reliable interpretation: the effect remains largely flat, indicating that typical debt levels do not meaningfully alter fraud risk after accounting for between-group differences.

6 Numerical Study

To truly interpret our results, we wanted to understand what type of underlying truth in a dataset would cause different model architectures to perform best. To accomplish this, we conducted a numerical study in which we simulated three different scenarios:

1. The underlying truth is that the response is a **linear** combination of the predictors
2. The underlying truth is that the response is a **non-linear** combination of the predictors
3. The underlying truth is that the response is a **nonlinear** combination of the predictors with additional **group effects**

6.1 Simulation procedure

To conduct this simulation, we first defined our predictors. For the purpose of this simulation we opted for only two predictors and generated them both from a standard normal distribution. Then, depending on the scenario, we generated η in different ways.

For the **linear scenario** we assumed the following formula: $\eta = -3 + 2x_1 - 1.5x_2$

For the **non-linear scenario**, η was a bit more complicated. We first define the following functions:

$$f_1(x) = 2\sin(4\pi x) \quad f_2(x) = -4(x - 0.25)^2 + 3e^{-40(x-0.75)^2}$$

And now η is defined as: $\eta = -3 + f_1(x_1) + f_2(x_2)$

For the **non-linear scenario with group effects**, we used the η formula from the previous scenario but now with additional random intercepts with the following formula (where G is the number of groups):

$$\eta = -3 + f_1(x_1) + f_2(x_2) + \text{re} \quad \text{re} = (re_1, \dots, re_G)^\top \sim \mathcal{N}(0, 2I_G)$$

We then plug our vector of η 's into the cumulative function for the logistic distribution, which returns us a vector of probabilities. Finally, we generate our response y from a binomial distribution with probabilities from our vector.

6.2 Scoring Model Performance

Each scenario imposes increasing modeling difficulty, and our goal is to assess the point at which simpler models (the GLM and GAM) begin to break down or lose reliability. To compare performance across scenarios, we evaluated two complementary metrics: the Area Under the Receiver Operating Characteristic Curve (AUC) and the Brier score. AUC measures a model's ability to discriminate between the positive and negative classes, with values near 1 indicating excellent separation and 0.5 corresponding to random guessing. The Brier score assesses calibration by quantifying the mean squared difference between predicted probabilities and the observed binary outcomes ^[3]; lower values indicate more accurate probabilistic predictions.

6.3 Analysis

Looking at Figure 5, we see how the AUC of each model changes with the increasing complexity of the underlying truth of the data. First, all models perform similarly on the linear data. Then, GLM appears to fall off in performance when moving to the slightly more complex non-linear data. As we then move to the final scenario with the non-linear data with random effects, we find that GAMM out-performs the other models with its more flexible structure. In Figure 6 we see a similar pattern. All models seem to perform similar on the purely linear data. On the non-linear data, we actually see a jump in performance for all models with GAM and GAMM slightly edging out GLM. Finally in the final scenario with non-linear data with random effects, we see a similar trend to what was shown in the AUC plot; furthering our understanding of a more complex model like GAMM being able to better capture these hidden relationships that the other less-complex models are unable to.

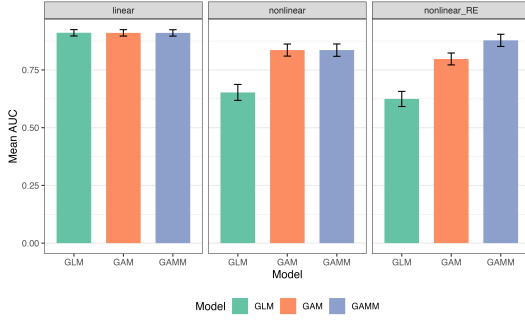


Figure 5: AUC by Model and Scenario

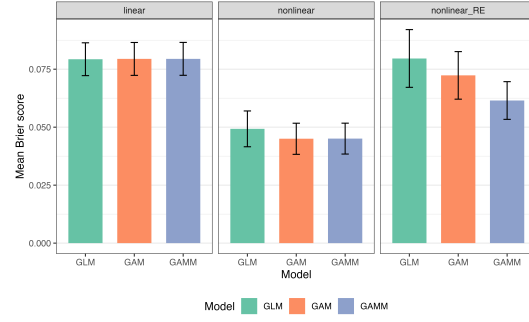


Figure 6: Brier score by Model and Scenario

7 Conclusion

7.1 Key Findings

Our investigation into credit card fraud suggests that accurate modeling requires a framework capable of accommodating both non-linear trend and group-level heterogeneity. While the GLM model provided a necessary baseline, our empirical analysis reveals its structural inadequacy in capturing the complex dynamics of fraud.

These empirical results were corroborated by our simulation study, which tested model performance against three distinct scenarios. We observed that while GLM model suffices for linear trends, its performance degrades in the presence of non-linearity. In scenario characterized by non-linear predictors and random group effects, the GAMM outperformed other models, achieving the highest AUC and the lowest Brier scores. This confirms that the superior performance of the GAMM on the observed data is not a result of overfitting, but a result of its structural alignment with hidden patterns of fraudulent behavior.

Beyond the structural comparison of model architectures, our analysis reveals critical determinants of fraud with direct implications for risk management strategies. We identified many significant variables, suggesting that risk is shaped by both transaction-level and customer-level factors. First, geography and merchant categories act as the most distinctive discriminators. The high variance in fraud rates across specific vendors suggests risk is clustered, necessitating step-up authentication and dynamic blacklisting. Second, our findings indicate a significant temporal component where fraud peaks in the fall and winter seasons, which implies banks should dynamically adjust model sensitivity during different periods. Finally, we also observed a distinct “U-shaped” risk curve regarding transaction amounts. Contrary to a linear relationship, fraud probability is elevated at both extremes: very low-value transactions (indicative of “probing” activity) and exceptionally high-value transactions carry the highest risk, while moderate amounts represent a relative “safe zone”.

7.2 Limitations

A critical limitation of this study is the handling of class imbalance. While estimation techniques rely on asymptotic consistency, the statistical power in binary classification is driven by the scarcity of the minority class rather than the total sample size.

The mean of a binary variable represents information content; consequently, in datasets where the event of interest is rare, the information available for maximum likelihood estimation is constrained not by the total sample size, but by the count of the minority class. Theoretical literature, specifically the work of King and Zeng, highlights that MLE of the logistic model suffers from small-sample bias that is inversely proportional to the frequency of events^[5]. Even in large samples, if the number of fraud cases is small, standard logit coefficients are systematically biased. This bias is predictable in direction: it leads to a consistent under-estimation of event probabilities. In the context of our study, this suggests that our probability estimates are likely conservative, potentially resulting in an inflated False Negative rate as the model optimizes for the majority class to minimize global error.

7.3 Areas for Future Study

To address the limitations imposed by rare-event data, future research can apply two primary methodological advancements.

First, we propose moving beyond standard Maximum Likelihood Estimation in favor of Penalized Likelihood, commonly known as the Firth method^[2]. This approach introduces a bias-correction term to the likelihood function, ensuring finite and consistent parameter estimates even in cases of complete or quasi-complete separation, which are common in sparse data. Unlike exact logistic regression, which is computationally prohibitive for large datasets, penalized likelihood incurs negligible additional computational cost.

Second, future work should evaluate data-level resampling techniques. Resampling techniques such as SMOTE (Synthetic Minority Over-sampling Technique) or strategic under-sampling of the majority class balance the class distribution a priori. By modifying the training data distribution before the model observes it, these methods enable the use of standard estimation techniques, avoiding the need for specialized corrections to the parameter estimates.

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